

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 9

Munhyeok Kang

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Topic

Orbital Calibration

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Orbital Calibration. State all assumptions and conventions.

$$\Delta\varpi = \beta \cdot 6\pi GM/[a(1-e^2)c^2]$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\beta = 1.84$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$\Delta\Omega = \alpha_3 f_3(a, e, i) + \alpha_4 f_4(a, e, i)$$

Show enough intermediate work that another student can reproduce the calculation.